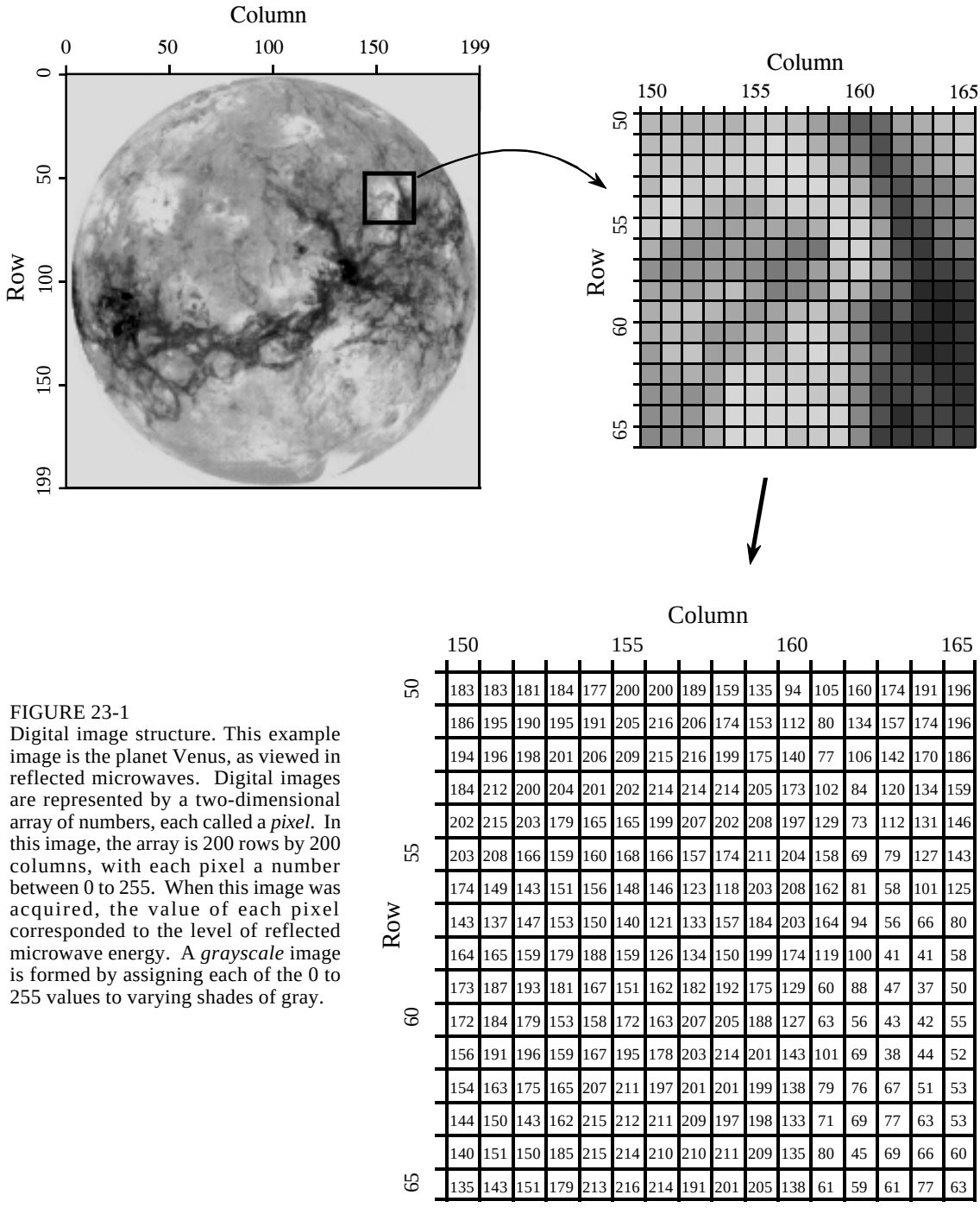

FIGURE 23-1
Digital image structure. This example image is the planet Venus, as viewed in reflected microwaves. Digital images are represented by a two-dimensional array of numbers, each called a *pixel.* In this image, the array is 200 rows by 200 columns, with each pixel a number between 0 to 255. When this image was acquired, the value of each pixel corresponded to the level of reflected microwave energy. A *grayscale* image is formed by assigning each of the 0 to 255 values to varying shades of gray.

probe takes samples every 10 meters along the planet's surface as it orbits overhead. This defines a square **sample spacing** and **sampling grid,** with each pixel representing a 10 meter by 10 meter area. Now, imagine what happens in a single microwave reflection measurement. The space probe emits